

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

***/ G. GOWTHAMI(192524285)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:G. GOWTHAMI

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

ANSWERS

1. Reactive Agent-Based Traffic Control System

Introduction

A **Reactive Agent** is an AI agent that responds immediately to changes in its environment. It does not require complex long-term planning. In a smart traffic control system, the agent continuously monitors traffic conditions and changes traffic signals according to real-time traffic information.

Objective

The main objectives are:

- Reduce traffic congestion.
- Minimize vehicle waiting time.
- Dynamically control traffic signals.
- Detect traffic jams and accidents quickly.
- Improve road utilization.
- Process real-time streaming data.

System Architecture

Traffic Sensors/Cameras → Streaming Platform → Data Processing → Reactive Agent → Traffic Signal Controller

Components

1. Traffic Sensors

- Detect vehicle count.
- Measure vehicle speed.

- Detect road occupancy.
- Identify abnormal traffic conditions.
- 2. **Streaming Data System**
 - Continuously receives sensor data.
 - Technologies such as **Apache Kafka** can be used.
- 3. **Stream Processing**
 - Processes incoming data in real time.
 - **Apache Spark Streaming** can calculate traffic density and vehicle flow.
- 4. **Reactive Agent**
 - Applies predefined rules.
 - Makes immediate decisions based on current traffic conditions.
- 5. **Signal Controller**
 - Changes red, yellow, and green signals according to the agent's decision.

Decision Logic

Example rules:

IF traffic_density > HIGH

THEN increase green-light duration

IF traffic_density < LOW

THEN decrease green-light duration

IF accident_detected = TRUE

THEN redirect traffic and alert authorities

IF emergency_vehicle_detected = TRUE

THEN provide green signal on emergency route

Streaming Data Processing

Traffic data continuously flows through the system:

Sensors → Kafka → Spark Streaming → Reactive Agent → Signal Controller

For example, if 100 vehicles are detected on Road A and only 20 vehicles on Road B, the agent can increase the green-light duration for Road A.

Advantages

- Fast response.
- Reduces congestion.
- Works with real-time data.
- Simple decision-making.
- Reduces unnecessary waiting.
- Supports emergency vehicles.

Conclusion

A reactive traffic-control agent provides **fast and automatic traffic management** by continuously observing traffic conditions and immediately responding through signal changes. Streaming technologies such as Kafka and Spark enable the system to process large volumes of real-time traffic data efficiently.

2. Goal-Based Agent for Healthcare Diagnosis

Introduction

A **Goal-Based Agent** makes decisions by considering a desired goal and selecting actions that help achieve that goal.

In healthcare, the goal may be:

Determine the most likely diagnosis and recommend an appropriate treatment pathway.

The system analyzes patient information and evaluates possible diagnostic options.

Objective

The healthcare agent aims to:

- Identify possible diseases.
- Analyze patient symptoms and medical history.
- Set diagnostic goals.
- Evaluate medical test results.
- Recommend suitable treatment pathways.
- Support doctors in decision-making.

System Architecture

Patient Data → Data Pipeline → Diagnostic Agent → Decision Engine → Diagnosis/Treatment Recommendation

Input Data

The agent can process:

- Patient symptoms.
- Age and gender.
- Medical history.
- Blood-test results.
- Imaging results.
- Vital signs.
- Medication history.
- Electronic health records.

Goal Formation

The agent establishes goals such as:

Goal 1: Identify possible disease.

Goal 2: Estimate disease probability.

Goal 3: Determine required tests.

Goal 4: Select appropriate treatment pathway.

Decision-Making

The agent compares patient information with medical knowledge and predictive models.

Example:

Symptoms:

Fever + cough + breathing difficulty

↓

Diagnostic Model

↓

Possible conditions:

Pneumonia

Influenza

Respiratory infection

↓

Evaluate test results

↓

Select most appropriate diagnostic pathway

Data Pipeline

A healthcare data pipeline can be:

Electronic Health Records → Data Ingestion → Data Cleaning → Feature Extraction → AI Model → Goal-Based Agent → Recommendation

Large healthcare datasets can be processed using distributed computing frameworks such as **Apache Spark**.

Example

Suppose a patient has:

- High temperature.
- Persistent cough.
- Low oxygen level.
- Abnormal chest X-ray.

The agent evaluates these inputs against its diagnostic models and determines which diagnostic pathway should be investigated further.

The final recommendation should be reviewed by a qualified healthcare professional rather than treated as an autonomous medical decision.

Advantages

- Faster analysis of large patient datasets.
- Consistent decision-making support.
- Helps prioritize diagnostic possibilities.
- Can integrate multiple data sources.
- Supports doctors with evidence-based recommendations.

Conclusion

A goal-based healthcare agent works toward specific diagnostic objectives. By combining patient data, data pipelines, medical knowledge, and decision-making models, it can provide useful **clinical decision support**.

3. Utility-Based Agent for E-Commerce Recommendations

Introduction

A **Utility-Based Agent** selects an action by calculating the usefulness or **utility** of different choices.

In e-commerce, the agent evaluates products and recommends the products that are most likely to satisfy a particular customer.

Objective

The system aims to:

- Understand customer preferences.
- Recommend relevant products.
- Maximize customer satisfaction.
- Improve product discovery.
- Increase recommendation quality.
- Process millions of customer interactions.

System Architecture

User Data → Data Processing → User Profile → Utility Calculation → Recommendation Agent → Recommended Products

Input Data

The system can analyze:

- Search history.
- Purchase history.
- Product views.
- Product ratings.
- Wishlist.
- Price preferences.
- Product categories.
- Similar-user behavior.

Utility Function

Each product receives a utility score.

For example:

$$U(p) = w_1R + w_2P + w_3S + w_4C$$

Where:

- **R** = relevance to user.
- **P** = price suitability.
- **S** = product popularity or quality.
- **C** = category preference.
- **w₁, w₂, w₃, w₄** = importance weights.

The system calculates utility for different products and ranks them.

Example

Suppose a customer searches for laptops.

Product	Relevance	Price	Rating	Utility
Laptop A	High	Medium	High	0.91
Laptop B	Medium	Low	High	0.82
Laptop C	High	High	Medium	0.76

The agent recommends **Laptop A** because it has the highest overall utility.

Large-Scale Data Processing

Millions of interactions can be processed using distributed technologies:

User Events → Kafka → Spark → Feature Processing → Recommendation Model → Utility Calculation → Ranking

The system can continuously update user profiles as new behavior is observed.

Advantages

- Personalized recommendations.
- Improved customer experience.
- Handles large datasets.
- Optimizes multiple factors simultaneously.
- Can adapt recommendations to changing preferences.

Conclusion

A utility-based recommendation agent evaluates multiple products using utility scores and selects the options that provide the greatest expected value to the user. Large-scale data processing allows the system to generate recommendations efficiently for millions of customers.

4. Learning Agent for Personalized Education System

Introduction

A **Learning Agent** improves its behavior based on experience and feedback.

In an AI-based education platform, the agent observes how students perform and continuously adapts the learning material to the student's ability.

Objective

The learning agent aims to:

- Understand student performance.
- Identify weak topics.
- Adapt content difficulty.
- Provide personalized exercises.
- Learn from student feedback.
- Improve recommendations over time.

System Architecture

Student Activity → Data Collection → PySpark Processing → Learner Model → Learning Agent → Personalized Content → Student Feedback

Input Data

The system collects:

- Quiz scores.
- Assignment results.
- Time spent on questions.
- Number of attempts.
- Wrong answers.
- Learning speed.
- Course completion.
- Student feedback.

Learning Process

The agent operates through a feedback loop:

Observe → Analyze → Decide → Teach → Receive Feedback → Learn → Adapt

For example:

Student takes mathematics quiz

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Score = 45%

↓

Agent identifies weak topic

↓

Provides easier exercises

↓

Student improves to 70%

↓

Agent increases difficulty

Use of PySpark

PySpark can process large-scale educational data.

Example:

Student Data

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PySpark Data Processing

↓

Performance Analysis

↓

Feature Generation

↓

Learning Model

↓

Personalized Recommendation

If a platform has millions of students, PySpark can distribute data processing across multiple machines.

Learning Components

A learning agent commonly contains:

1. Performance Element

Chooses what content should be presented.

2. Learning Element

Learns from student performance.

3. Critic

Evaluates whether the selected learning strategy was successful.

4. Problem Generator

Creates new learning activities or exercises.

Advantages

- Personalized learning.
- Continuous improvement.
- Automatic difficulty adjustment.
- Scalable to many students.
- Identifies learning gaps.
- Uses feedback to improve recommendations.

Conclusion

A learning agent makes education more adaptive by continuously learning from student behavior. With distributed processing frameworks such as PySpark, educational platforms can process large datasets and provide personalized learning experiences at scale.

5. Multi-Agent System for Disaster Management

Introduction

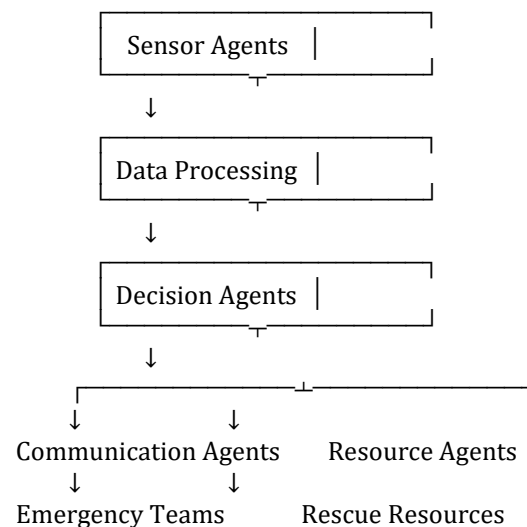
A **Multi-Agent System (MAS)** consists of multiple intelligent agents that work together to solve a complex problem. In disaster management, different agents can monitor sensors, analyze conditions, coordinate resources, and communicate emergency information.

Objective

The system aims to:

- Detect disasters quickly.
- Analyze real-time sensor data.
- Coordinate emergency resources.
- Identify dangerous areas.
- Communicate with rescue teams.
- Support real-time decision-making.

System Architecture



Types of Agents

1. Sensor Agents

Collect information from:

- Temperature sensors.
- Water-level sensors.
- Smoke detectors.

- Cameras.
- Seismic sensors.
- Weather stations.

2. Decision Agents

Analyze incoming information and determine:

- Disaster severity.
- Affected areas.
- Evacuation requirements.
- Priority locations.
- Required emergency resources.

3. Communication Agents

Send information to:

- Emergency services.
- Hospitals.
- Police.
- Fire departments.
- Rescue teams.
- Public warning systems.

4. Resource Agents

Track:

- Ambulances.
- Rescue vehicles.
- Medical supplies.
- Rescue personnel.
- Shelters.

Coordination Strategy

Agents communicate with each other and share information.

For example:

Sensor Agent

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Detects flood

↓

Decision Agent

↓

Determines severity

↓

Communication Agent

↓

Alerts authorities

↓

Resource Agent

↓

Dispatches rescue teams

Real-Time Data Processing

The system can use a streaming architecture:

IoT Sensors → Kafka → Spark Streaming → Agent Network → Decision → Emergency Response

Streaming processing allows agents to react to changing disaster conditions quickly.

Example

During a flood:

1. Water-level sensors detect rapidly increasing water levels.
2. Sensor agents transmit the data.
3. Decision agents classify the situation as high risk.
4. Communication agents send alerts.
5. Resource agents identify available rescue vehicles.
6. Rescue teams are dispatched.
7. Agents continue monitoring the situation and update decisions.

Advantages

- Distributed decision-making.
- Fast disaster detection.
- Parallel processing.

- Better resource coordination.
- Fault tolerance.
- Real-time response.
- Scalable architecture.

Conclusion

A multi-agent disaster-management system combines multiple specialized agents to detect disasters, share information, make decisions, and coordinate emergency responses. It is particularly useful for complex environments where decisions must be made quickly using continuously changing data.

Overall Comparison

Agent Type	Application	Main Decision Principle
Reactive Agent	Smart Traffic Control	Responds immediately to current conditions
Goal-Based Agent	Healthcare Diagnosis	Chooses actions to achieve a diagnostic goal
Utility-Based Agent	E-Commerce	Chooses the option with the highest utility
Learning Agent	Personalized Education	Improves through feedback and experience
Multi-Agent System	Disaster Management	Multiple agents cooperate to solve a complex problem

Final Conclusion

These five systems demonstrate different approaches to **AI agent-based decision-making**:

- **Reactive agents** are best when fast responses to changing environments are required.
- **Goal-based agents** are useful when the system must work toward a specific objective.
- **Utility-based agents** are suitable when several alternatives must be compared and optimized.
- **Learning agents** improve their performance through experience and feedback.
- **Multi-agent systems** are useful when multiple intelligent components need to collaborate on a complex real-time problem.

Together, they demonstrate how **AI agents, streaming data processing, distributed computing, and intelligent decision-making** can be combined to build modern real-world systems.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness